

Ex 1.

1. No. 要单独一个 a , 要 m 个 $+$ 或 $*$ 结尾

2. $S \Rightarrow SS^*$

$\Rightarrow SS^*S^*$

$\Rightarrow aS^*S^*$

$\Rightarrow aa^*S^*$

$\Rightarrow aa^*SS^+^*$

$\Rightarrow aa^*aS^+^*$

$\Rightarrow aa^*aa^+^*$

3. $S \Rightarrow SS^*$

$\Rightarrow SSS^+^*$

$\Rightarrow SSSa^+^*$

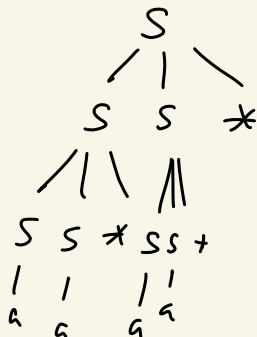
$\Rightarrow Saa^+^*$

$\Rightarrow SS^+^*aa^+^*$

$\Rightarrow Sa^+^*aa^+^*$

$\Rightarrow aa^+^*aa^+^*$

4.



$$5. \quad S \rightarrow \underline{S}S+ \\ \quad \quad \quad | \underline{S}S* \\ \quad \quad \quad | a$$

$$\{ S \rightarrow a S' \}$$

$$S' \rightarrow \underline{S}+S' \mid \underline{S}*S' \mid \varepsilon \quad (\text{循环引用})$$

Ex2

$$S \rightarrow a \overset{\vee}{B}$$

$$B \rightarrow S+B \mid \varepsilon$$

$$1. \quad \text{FIRST}(S) = \{a\} \quad \text{FOLLOW}(S) = \{+, \$\}$$

$$\text{FIRST}(B) = \{a, \varepsilon\} \rightarrow \text{FOLLOW}(B) = \{+, \$\}$$

$$\text{FIRST}(aB) = \{a\}$$

$$\text{FIRST}(S+B) = \{a\}$$

$$\text{FIRST}(\varepsilon) = \{\varepsilon\}$$

	a	+	\$
S	$S \rightarrow aB$		
B	$B \rightarrow S+B$	$B \rightarrow \varepsilon$	$B \rightarrow \varepsilon$

$$2. \quad \text{Yes.} \quad \text{FIRST}(S+B) \cap \text{FIRST}(\varepsilon) = \emptyset$$

$$\text{FIRST}(S+B) \cap \text{FOLLOW}(B) = \emptyset$$

3.

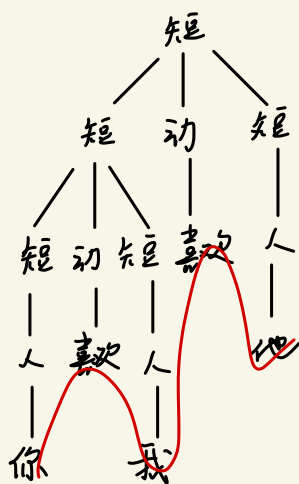
MATCHED	Stack	INPUT	ACTION
	<u>S</u> \$	<u>aaaa</u> + + + \$	
	aB \$	aaaa + + + \$	Output <u>S → aB</u>
a	<u>B</u> \$	<u>aaa</u> + + + \$	Match a
a	S + B \$	aaa + + + \$	Output B → S + B
a	aB + B \$	aaa + + + \$	Output <u>S → aB</u>
aa	B + B \$	aa + + + \$	Match a
aa	S + B + B \$	aa + + + \$	Output B → S + B
aa	aB + B + B \$	aa + + + \$	Output <u>S → aB</u>
aaa	B + B + B \$	aa + + + \$	Match a
aaa	S + B + B + B \$	aa + + + \$	Output B → S + B
aaa	aB + B + B + B \$	aa + + + \$	Output S → aB
aaaa	B + B + B + B \$	++ + \$	Match a
aaaa	+ B + B + B \$	++ + \$	Output B → ε
aaaa +	B + B + B \$	++ \$	Output B → ε Match \$ +
aaaa +	+ B + B \$	++ \$	Output B → ε <u>Output B → ε</u>
aaaa + +	B + B \$	+ \$	Match \$ +
aaaa + + +	+ B \$	+ \$	<u>Output B → ε</u>
aaaa + + +	B \$	\$	Match \$ +
aaaa + + +	\$	\$	<u>Output B → ε</u>
			Accept. 栈顶为 \$

FIRST (预测) = {你, 我, 他} →

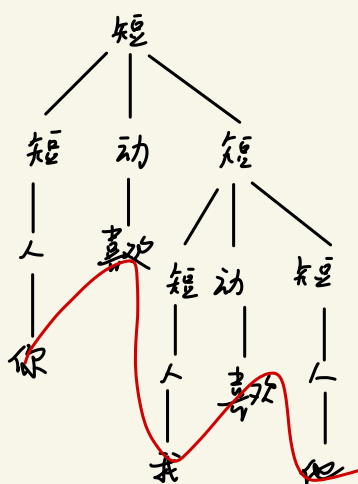
Bonus

Ex1

有歧义. 比如“你喜欢我喜欢他”有如下两种语法树



(你喜欢我) 喜欢他
↓
短



你喜欢 (我喜欢他)